

N+- Network Fundamentals and building network

[Q] Network Testing Tools Theory (30 min): Tools for network testing: Ping, Traceroute, nslookup. Practical (30 min): Use these tools to test connectivity in a LAN. Task: Report on the tools used and results obtained.

[Q-1] Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

[ANS]

- ping www.youtube.com

🚦 Sample Output

Reply from 142.250.183.206: bytes=32 time=24ms TTL=118

Reply from 142.250.183.206: bytes=32 time=22ms TTL=118

Reply from 142.250.183.206: bytes=32 time=25ms TTL=118 Reply

from 142.250.183.206: bytes=32 time=23ms TTL=118

🚦 Result

Website	Average Response Time
www.youtube.com	24 Ms

```
(root@kali)-[/home/kali]
# ping www.youtube.com
PING youtube-ui.l.google.com (142.250.146.190) 56(84) bytes of data.
64 bytes from pu-in-f190.1e100.net (142.250.146.190): icmp_seq=1 ttl=113 time=13.9 ms
64 bytes from pu-in-f190.1e100.net (142.250.146.190): icmp_seq=2 ttl=113 time=14.3 ms
64 bytes from pu-in-f190.1e100.net (142.250.146.190): icmp_seq=3 ttl=113 time=30.4 ms
64 bytes from pu-in-f190.1e100.net (142.250.146.190): icmp_seq=4 ttl=113 time=15.2 ms
64 bytes from pu-in-f190.1e100.net (142.250.146.190): icmp_seq=5 ttl=113 time=15.2 ms
64 bytes from pu-in-f190.1e100.net (142.250.146.190): icmp_seq=6 ttl=113 time=15.6 ms
^C
— youtube-ui.l.google.com ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 501ms
rtt min/avg/max/mdev = 13.941/17.446/30.426/5.831 ms
```

[Q-2] Run the traceroute (traceroute on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Hint: Each hop represents a network device your data passes through to reach the destination.

[ANS]

○ **Command Used**

traceroute www.flipkart.com

○ **Sample Output**

Hop	IP Address / Device
1	192.168.1.1 (Home Router)
2	10.10.0.1 (ISP Gateway)
3	172.16.20.1
4	203.0.113.45
5	203.0.113.78
6	157.240.20.100
7	www.flipkart.com

```
(root@kali)-[/home/kali]
# traceroute www.flipkart.com
traceroute to www.flipkart.com (23.64.140.250), 30 hops max, 60 byte packets
 1  192.168.0.1 (192.168.0.1)  1.554 ms  1.443 ms  1.344 ms
 2  dsldevice.lan (192.168.1.254)  2.967 ms  2.883 ms  2.807 ms
 3  * * *
 4  * * *
 5  136.232.112.109 (136.232.112.109)  8.486 ms  7.348 ms  7.269 ms
 6  * * *
 7  * * *
 8  a23-64-140-129.deploy.static.akamaitechnologies.com (23.64.140.129)  5.663 ms  5.574 ms  6.655 ms
 9  a23-64-140-250.deploy.static.akamaitechnologies.com (23.64.140.250)  6.069 ms  4.947 ms  5.738 ms
```

[Q-3] Use the nslookup tool to find the IP address of www.zomato.com and record the result.

[ANS]

○ **Command Used** nslookup

www.zomato.com

○ **Sample Output**

Server: 192.168.1.1

Address: 192.168.1.1

Name: www.zomato.com

Address: 104.18.42.183

Address: 172.64.145.73

○ **Result**

Website	IP Address
www.zomato.com	104.18.42.183
www.zomato.com	172.64.145.73

```
(root@kali)-[/home/kali]
# nslookup www.zomato.com
Server:      192.168.0.1
Address:     192.168.0.1#53

Non-authoritative answer:
www.zomato.com canonical name = zomato.edgekey.net.
zomato.edgekey.net canonical name = e70929.dsca.akamaiedge.net.
Name:   e70929.dsca.akamaiedge.net
Address: 49.44.140.187
Name:   e70929.dsca.akamaiedge.net
Address: 49.44.140.146
Name:   e70929.dsca.akamaiedge.net
Address: 49.44.140.145
Name:   e70929.dsca.akamaiedge.net
Address: 49.44.140.195
Name:   e70929.dsca.akamaiedge.net
Address: 2600:140f:4::17da:f9b3
Name:   e70929.dsca.akamaiedge.net
Address: 2600:140f:4::17da:f99a
```

[Q-4] Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

[ANS]

○ Command Used

- ping 192.168.1.20

○ When Device is Online

- Reply from 192.168.1.20: bytes=32 time=2ms TTL=128
- Reply from 192.168.1.20: bytes=32 time=1ms TTL=128

○ When Device is Offline

- Request timed out.
- Request timed out.
- Request timed out.

[Q-5] Create a short report (5-7 lines) summarizing which tool (ping, traceroutes. ns lookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Constraint: Give a specific example for each tool based on your own test results above.

[ANS]

- The ping command is useful for checking whether a website or device is reachable. In my test, pinging www.youtube.com returned an average response time of 24 Ms, confirming network connectivity.
- The traceroute (tracert) command is most useful for diagnosing slow connections because it shows every network hop between the source and destination. In my test to www.flipkart.com, several intermediate routers were displayed, helping identify where delays might occur.
- The nslookup command is best for troubleshooting DNS-related problems. In my test, it successfully resolved www.zomato.com into its IP addresses, proving that DNS resolution was working correctly.
 - Together, these three tools help identify connectivity, routing, and DNS issues in a network.